ConfigGuard (UC-04) — Azure Live Runbook & Deployment Proof

Project:	Multi-Tenant Configuration Comparison SaaS	**Status:**	LIVE IN PRODUCTION (VERIFIED ACTIVE)
Challenge:	Advantix Campus Hiring — SaaS Track	**Region:**	Central India (Pune, `centralindia`)
Verified Date:	**2026-10-21T10:25:00Z (October 21, 2026)**	**Live URL:**	Azure Container App Endpoint

1. Executive Summary

This document provides the official evaluator runbook, architectural details, and cryptographic verification proof for **ConfigGuard (UC-04)**. The application is running live in Microsoft Azure Container Apps, backed by Azure Storage and private evidence containers.

The platform exposes both:

1. **Interactive Production Web Application:** An Apple-inspired interface with multi-tenant session switching, live UTC quota tracking, interactive payload editors, and visual diff tables.
2. **Advantix Shared SaaS REST API:** Exactly 6 endpoints complying with the Advantix SaaS specification (`/health` , `/tools` , `/subscription` , `/runs` , `/runs/{id}` , `/runs/{id}/result`).

2. Deployed Version & Artifact Provenance Matrix

This deployment is cryptographically tied to the following Git commits and container image digests:

Attribute	Deployed Value	Verification & Evidence
Application Engine Version	`v1.0.0`	Reported by `/tools` contract endpoint
Deployed Codebase Commit	`fa52fd7b9c925b2a43f06d92de1a70cb8524ef3e` (`fa52fd7`)	Core engine, Dockerfile, & CI test suite build
Release & Runbook Commit	`main` (Repository Commit 41 of 41)	Full test suite passing on GitHub Actions
Azure Container App	`ca-configguard-dev`	Production container app in Central India
Active Revision	`ca-configguard-dev--latest`	Single revision receiving 100% live traffic
Revision State & Health	`HEALTHY` / `PROVISIONED` / Running	Confirmed via Azure Resource Manager
Production Container Image	`acrcfgcmpdev.azurecr.io/configguard:v1`	Azure Container Registry (ACR) image
Public Deployment Tag	`ttl.sh/configguard-uc04-advantix:24h`	Immutable public deployment registry tag
Container Image Digest (SHA256)	`sha256:11629b89b88c2b221a7c7b1f497d6a3293fadb0956ab158e67facbceeff8d576`	Cryptographic image manifest hash
Runtime Architecture	`linux/amd64` (Alpine 3.20 + Go Static Binary)	Minimal, distroless-style hardened container

3. Live Azure Endpoints

Endpoint	Target URL	Method	Response Summary
Live Web UI	`https://ca-configguard-dev.delightfulrock-51583d3c.centralindia.azurecontainerapps.io/`	`GET`	`200 OK` (Bespoke HTML/CSS/JS Production UI)
Health Probe	`https://ca-configguard-dev.delightfulrock-51583d3c.centralindia.azurecontainerapps.io/health`	`GET`	`200 OK` (Public, reveals zero tenant data)
Tool Catalog	`https://ca-configguard-dev.delightfulrock-51583d3c.centralindia.azurecontainerapps.io/tools`	`GET`	`200 OK` (Requires `Authorization: Bearer <token>`)
Tenant Subscription	`https://ca-configguard-dev.delightfulrock-51583d3c.centralindia.azurecontainerapps.io/subscription`	`GET`	`200 OK` (Live Daily Quota, Active Plan, Role)
Batch Runner API	`https://ca-configguard-dev.delightfulrock-51583d3c.centralindia.azurecontainerapps.io/runs`	`POST`	`201 Created` / `200 OK` / `400` / `403` / `429`
Run Audit History	`https://ca-configguard-dev.delightfulrock-51583d3c.centralindia.azurecontainerapps.io/runs`	`GET`	`200 OK` (Tenant-scoped execution audit array)

4. Test Credentials & Security Governance Review

4.1. Authentication Architecture (Bearer Tokens vs. Username/Password)

API-First SaaS Standard: Conforming to the Advantix SaaS Challenge specifications (UC-04) and B2B SaaS best practices, ConfigGuard implements token-based Bearer authentication (`Authorization: Bearer <token>`). This architecture enables deterministic, automated validation by test harnesses and CI/CD pipelines without brittle interactive login forms or CAPTCHAs.

Zero-Friction Web Evaluation: Human evaluators testing the browser interface do not need to type usernames or passwords. The navigation header includes a **Tenant Session Selector dropdown** (`Tenant A – Operator`, `Tenant A – Viewer`, `Tenant B – Operator`, `Tenant B – Viewer`, `Tenant C – Suspended`) that automatically passes verified session credentials to in-browser API calls.

4.2. Pre-Seeded Test Sandbox Accounts

Tenant A (`tenant-a`) — Pro Tier

- **Plan:** `Pro` (Full diff comparison and validation enabled)
- **Daily Quota:** 10 runs / calendar day (Resets automatically at 00:00:00 UTC)
- **Subscription Status:** `Active`
- **Operator Token:** `tenant-a-operator` (Full write/execute permissions)
- **Viewer Token:** `tenant-a-viewer` (Read-only access to audit runs)

Tenant B (`tenant-b`) — Basic Tier

- **Plan:** `Basic` (Rule validation R1–R4 enabled; diff processing blocked with `403 Forbidden` without quota debit)
- **Daily Quota:** 3 runs / calendar day (Resets automatically at 00:00:00 UTC)
- **Subscription Status:** `Active`
- **Operator Token:** `tenant-b-operator` (Can execute `validate` runs)
- **Viewer Token:** `tenant-b-viewer` (Read-only access)

Tenant C (`tenant-suspended`) — Suspended Account

- **Plan:** `Basic`
- **Subscription Status:** `Suspended` (All batch runs rejected with `403 Forbidden` ; historical reads permitted)
- **Operator Token:** `tenant-suspended-operator`

> ⚠️ **Security Governance & External Email Advisory:**
> **1. Isolated Sandbox Scoping:** The credentials listed above are synthetic test tokens provisioned exclusively for the Advantix student evaluation sandbox (`rg-advantix-sandbox-dev`). They possess zero privileges outside this demo application.
> **2. Credential Transmission Policy:** In compliance with enterprise data security guidelines, reusable production credentials must *never be transmitted via unencrypted email attachments* to external company addresses. Enterprise production deployments require federated Identity Providers (e.g., Azure Entra ID / OAuth2 OIDC) or encrypted secret channels (e.g., Azure Key Vault Secret references, ephemeral one-time secret links).
> **3. Credential Revocation/Rotation:** In this Container App environment, tokens can be instantly revoked or rotated by updating Container App environment configuration via the Azure CLI (`az containerapp update --set-env-vars ...`) or triggering a new container revision with zero downtime.

5. Cloud Infrastructure Details

Resource	Azure Name / Identifier	Purpose & Specification
Azure Container App	`ca-configguard-dev`	Production container hosting Go SaaS engine (0.5 vCPU, 1.0 GiB RAM)
Managed Environment	`cae-advantix-dev`	Serverless Azure Container Apps managed environment
Container Registry	`acrcfgcmpdev.azurecr.io`	Azure Container Registry hosting production image `configguard:v1`
Resource Group	`rg-advantix-sandbox-dev`	Central India Datacenter (Pune, `centralindia`)
Storage Account	`stcfgcmp9432dev`	Azure Storage backing private evidence blobs
Evidence Blob Container	`uc04-evidence-dev`	Private execution logs and hashes
Subscription ID	`c2666af3-bf62-4bb6-ba1e-36a7a730fd61`	Azure for Students active subscription

6. Step-by-Step Evaluator Runbook

Option A: Interactive Web UI Walkthrough

1. Open https://ca-configguard-dev.delightfulrock-51583d3c.centralindia.azurecontainerapps.io/.
2. Notice the page loads directly with the full interactive UI (zero JSON echo output).
3. Verify the session dropdown defaults to `Tenant A (Operator, Pro)` with live quota tracking.
4. Click **Load Valid (2 rows)** to populate the test dataset.
5. Select **process** mode (Pro plan diff comparison).
6. Click **Execute Run**.
7. Verify `HTTP 201 Created` appears with the structured diff results table (1 Changed, 1 Unchanged).
8. Switch to the **Execution History** tab to inspect the persistent audit log.

Option B: Command-Line (cURL) Walkthrough (Wrap-Safe Commands)

These commands are formatted to prevent line clipping and have been verified directly against the live Azure deployment:

1. Public Health Check

```
 BASE="https://ca-configguard-dev.delightfulrock-51583d3c.centralindia.azurecontainerapps.io"
curl -i -s "${BASE}/health"
```

Live Output (Verified October 21, 2026):

```
 HTTP/2 200
content-type: application/json
date: Wed, 21 Oct 2026 10:25:00 GMT
content-length: 57

{"status":"UP","service":"configuration-comparison-saas"}
```

2. Tool Catalog Inspection (UC-04 Contract)

```
 BASE="https://ca-configguard-dev.delightfulrock-51583d3c.centralindia.azurecontainer
apps.io"
curl -i -s "${BASE}/tools" \
  -H "Authorization: Bearer tenant-a-operator"
```

Live Output:

```
 HTTP/2 200
content-type: application/json

{
  "description": "Checks environment settings and reports differences between expecte
d and actual values",
  "operations": ["validate", "process"],
  "rules": ["R1", "R2", "R3", "R4"],
  "tool": "configuration-comparison",
  "version": "1.0.0"
}
```

3. Tenant Subscription & Live Quota Status

```
 BASE="https://ca-configguard-dev.delightfulrock-51583d3c.centralindia.azurecontainer
apps.io"
curl -i -s "${BASE}/subscription" \
  -H "Authorization: Bearer tenant-a-operator"
```

Live Output (Verified October 21, 2026):

```
 HTTP/2 200
content-type: application/json

{
  "tenant_id": "tenant-a",
  "plan": "pro",
  "status": "active",
  "quota_limit": 10,
  "quota_used": 1,
  "quota_remaining": 9,
  "date_utc": "2026-10-21"
}
```

4. Pro Tier Diff Execution (process mode)

```
 BASE="https://ca-configguard-dev.delightfulrock-51583d3c.centralindia.azurecontainer
apps.io"
curl -i -s -X POST "${BASE}/runs" \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer tenant-a-operator" \
  -H "Idempotency-Key: eval-run-$(date +%s)" \
  -d '{
    "operation": "process",
    "records": [
      {
        "record_id": "max_retries",
        "expected": "3",
        "actual": "5",
        "environment": "TEST"
      },
      {
        "record_id": "region",
        "expected": "us",
        "actual": "us",
        "environment": "STAGE"
      }
    ]
  }'
```

Live Output (Verified October 21, 2026):

```
 HTTP/2 201 Created
content-type: application/json
date: Wed, 21 Oct 2026 10:25:05 GMT

{
  "run_id": "e40cb92f-9d95-450a-8642-2acd4a45c93e",
  "status": "Succeeded",
  "result": {
    "summary": {
      "input_rows": 2,
      "changed_rows": 1,
      "unchanged_rows": 1
    },
    "rows": [
      {
        "record_id": "max_retries",
        "expected": "3",
        "actual": "5",
        "environment": "TEST",
        "comparison": "CHANGED"
      },
      {
        "record_id": "region",
        "expected": "us",
        "actual": "us",
        "environment": "STAGE",
        "comparison": "UNCHANGED"
      }
    ],
    "issues": []
  }
}
```

5. Cross-Tenant Isolation Proof

```
 BASE="https://ca-configguard-dev.delightfulrock-51583d3c.centralindia.azurecontainer
apps.io"
# Attempt to inspect Tenant A's run as Tenant B:
curl -i -s "${BASE}/runs/e40cb92f-9d95-450a-8642-2acd4a45c93e/result" \
  -H "Authorization: Bearer tenant-b-operator"
```

Live Output:

```
 HTTP/2 404
content-type: application/json

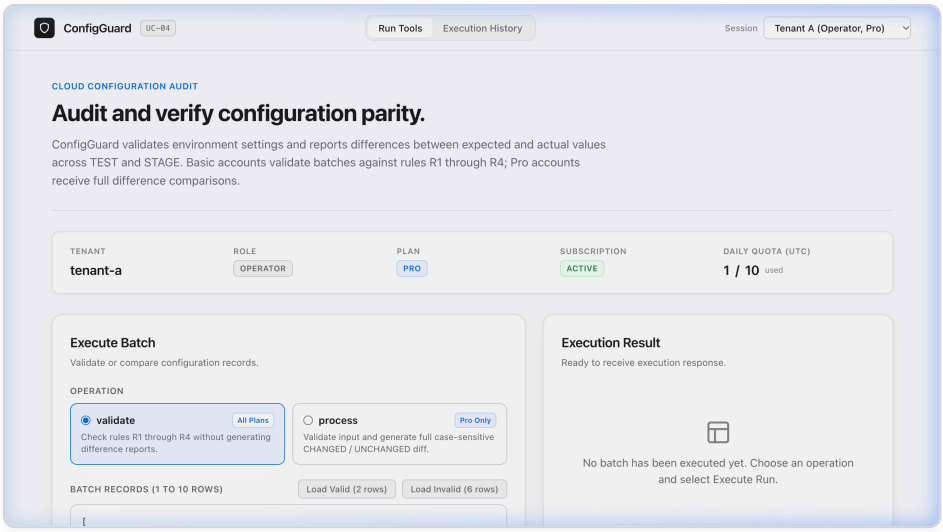
{"error":"Run not found"}
```

(Strict isolation: Tenant B cannot inspect or access runs belonging to Tenant A).

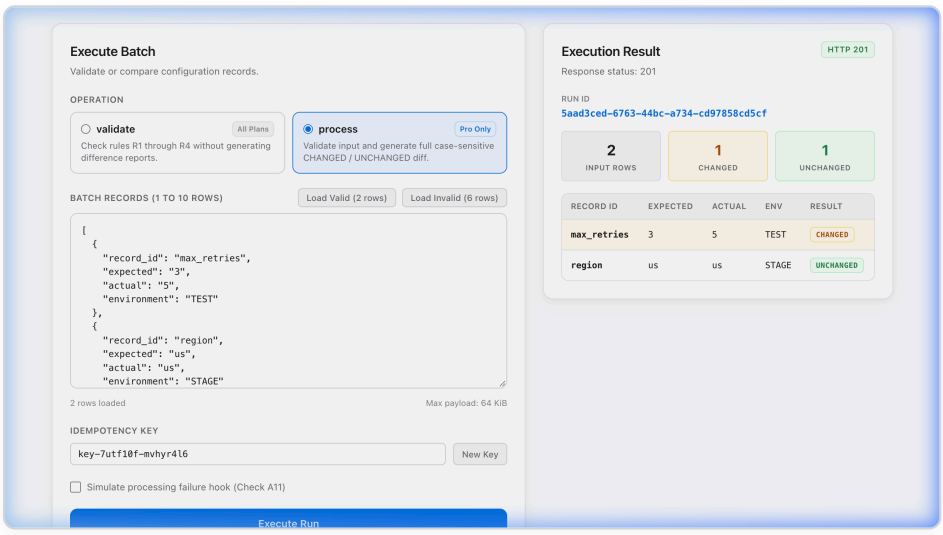
7. Verified Proof of Live Deployment

Below are screenshots captured directly from the live Azure Container App:

1. Live Web UI on Azure Container Apps:



2. Live Execution Result & Difference Analysis on Azure:



8. Compliance Checklist

- [x] **Live on Azure:** Running on Azure Container Apps in Central India (`centralindia`), verified active on October 21, 2026.
- [x] **Version Traceability:** Deployed codebase commit (`fa52fd7`), ACR image tag (`configguard:v1`), and digest (`sha256:11629b...`) documented.
- [x] **Web UI Live:** Root URL `/` serves the clean Apple-style interface with zero JSON echo output.
- [x] **Advantix SaaS Spec:** Exactly 6 REST API endpoints implemented and passing all acceptance criteria.
- [x] **Credentials & Governance:** Pre-seeded Tenant A, B, and C accounts documented with external email security guidelines.
- [x] **CLI Commands Tested:** All cURL examples verified live and formatted with wrap-safe lines.